GSoC 2027 Proposal: Reliable, CSI Producing OFDM Receiver for GNU Radio

Refined proposal (ECE 310 Milestone 4, Team G1). Organisation: GNU Radio

Skills	Python, C++, DSP (OFDM, synchronisation, channel estimation), GNU Radio blocks, unit testing
Applicant	One applicant per proposal (GSoC is individual). Each member who applies copies this proposal and fills in Section 10.
Potential mentors	Ask on the GNU Radio Matrix chat / Discourse before naming anyone. Candidates: gr-digital reviewers and the maintainers active on #6891 and #8271.
Before submitting	Check the official GNU Radio GSoC 2027 ideas list; align with a listed idea or get a mentor's OK for this one.

1. Synopsis

GNU Radio's generic OFDM transmitter/receiver (`gr-digital`) is where students, researchers and many Wi-Fi-sensing experiments start. Our measurements show that it fails **silently** in three ways: (1) about 4 % of back-to-back packets are dropped on a perfect channel and up to 11.5 % on multipath (#6891); (2) flowgraphs freeze without an exception when a frame does not fit a block's buffer (the symptom in #7997); (3) packets of 1020 bytes or more are never transmitted, and the flowgraph ends normally with zero output. The receiver also computes an accurate per packet channel estimate (CSI) but does not expose it.

This project will: **(a)** fix the multi-packet drop **in the receiver**; **(b)** turn the silent failures into clear errors or graceful drops, with tests; and **(c)** add a supported per-packet CSI output (with packet number and timestamp) plus an example that writes a CSI dataset from a configurable multipath channel. Together, these make GNU Radio a reliable, open CSI source for Wi-Fi-sensing research.

2. What changed since the M3 draft (and why)

M3 draft	Refined (M4)	Reason
Three receiver options, TX rolloff as a fallback "documented default"	**Receiver fix only.** The TX-rolloff fallback is removed	Maintainer review of PR #8271: changing the rolloff "hides a bug". A TX default does not help users whose bursts come from other transmitters
Test PR in weeks 1–2	The multi-packet test is already submitted in our one pull request, PR #8271 (open). A revision with `expectedFailure` is prepared and will be pushed to the same PR after the maintainer confirms. Weeks 1–2 build the full benchmark matrix	Gives the fix a ready acceptance test. Removing the `expectedFailure` decorator becomes the measurable goal at midterm

M3 draft	Refined (M4)	Reason
Prior-work links missing	Filled in (Section 4)	Our issue comment on #6891 and our one pull request (#8271) now exist
Generic risks	Risks tied to the actual review feedback	Section 8

3. Benefits to the community

Every user of `ofdm_loopback.grc` and `ofdm_tx` / `ofdm_rx` gets a receiver that does not silently lose packets, whatever the transmitter does.

Clear errors replace hangs that currently cost users hours (#7997).

Regression tests for multi-packet operation, large frames and buffer limits protect future changes, including the GNU Radio 4.0 port.

A per-packet CSI output opens GNU Radio to the Wi-Fi-sensing community, which today depends on closed chip-specific CSI tools.

4. Evidence of prior work (ECE 310 project, 2026)

Item	Link / location
Measurements on #6891	https://github.com/gnuradio/gnuradio/issues/6891#issuecomment 6787055390
Our only pull request so far: first multi-packet OFDM QA test	https://github.com/gnuradio/gnuradio/pull/8271 (submitted 11 Nov 2026; open; maintainer asked a question on 12 Nov; revision prepared, not pushed yet)
Scheduler ruled out (streaming = offline, 5/5 seeds)	M2 report, team repository
Sample-level model: packet lost iff its trigger is < frame_len − 1 samples after the last used trigger; predicts every loss in 100/100 runs	M3/M4 report, `e8_trigger_timing.py` , `e8b_length_scan.py`
TX workaround evaluated over 4 multipath profiles × 5 SNRs, CFO ≤ 3.5 subcarriers, 2 GNU Radio builds	M3/M4 report Section 6.4
CSI from `ofdm_chanest_vcvc` : −19 dB NMSE at 15 dB SNR, −34 dB at 30 dB; 7–9 % CSI samples missing; 7–8 % sensing bias	M3/M4 report Section 6.5
Freeze and ≥ 1020 B refusal reproduced	M3/M4 report Section 6.6
Team repository	https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing

5. Technical plan

5.1 Receiver-side fix for multi-packet drops (core)

Current behaviour. `header_payload_demux` ignores a trigger that falls inside the frame it is consuming. With touching bursts, the Schmidl & Cox plateau trigger lands 69–73 samples after the true burst start, so a trigger 1–4 samples early loses the packet.

Options to prototype and compare (decision with mentors by week 4):

1. **Early-trigger tolerance** in `header_payload_demux` : accept a trigger that falls in the last k samples of the current frame and re-align the next frame start. Smallest change; must not break users who rely on the current behaviour (default k to be agreed).
2. **Fine timing refinement** after detection: cross-correlate with the known second sync word to place the trigger consistently, before the demux. Keeps the demux unchanged; costs CPU.

Acceptance criteria. With `rolloff=0` at the TX: PDR = 100 % on the test matrix (lengths 20– 1000 B; flat, mild, indoor and harsh multipath; CFO ≤ 3.5 subcarriers; SNR ≥ 25 dB); the `expectedFailure` decorator on the #8271 test can be removed; no regression in existing QA tests; CPU cost measured and reported.

5.2 Loud failures instead of silent ones

`header_payload_demux` : a maximum-payload-length check (the source notes "There is currently no 'max payload len' param"). Drop the frame and log a warning instead of killing the thread.

Tagged-stream blocks: when a packet cannot fit the buffer, raise a clear error naming the block, the requested and available sizes and the fix (`set_min_output_buffer`), and stop the flowgraph instead of hanging.

≥ 1020 B TX refusal (`tagged_stream_mux` , max 8191 items available): fix the buffer sizing or document a hard limit with an explicit error. Report as a new issue first, with our reproducer. Tests that assert an error or warning instead of a hang, with a watchdog so CI never blocks.

5.3 Per-packet CSI output

Optional message port on `ofdm_rx` that emits, for every detected frame, a PMT dict: {packet_num, sample_offset, carrier_offset, csi (c32vector over occupied carriers), header_ok, crc_ok} .

Example `ofdm_csi_dataset.py` / `.grc` : TX → configurable multipath + Doppler channel → RX → CSI file in a documented NumPy/HDF5 layout compatible with SenseFi-style pipelines. Validation: CSI vs ground-truth |H(f)| NMSE vs SNR (as in our E9). Lost packets are reported explicitly, so sensing pipelines can use correct timestamps.

5.4 Documentation

A tutorial page, "OFDM packet reliability and CSI in GNU Radio", covering the measurement method (sequence-numbered packets, watchdog, trigger logging, replay model) and the new CSI output.

6. Timeline and milestones (12 weeks + community bonding)

Everything in this table is **planned work for GSoC 2027**. None of it has been done or submitted yet. So far the

team has submitted one pull request (#8271) and one issue comment (#6891).

Period	Work	Deliverable / milestone
Community bonding	Agree design options with mentors; source build of `main`; finish the review of our existing PR #8271	Design note
Weeks 1–2	Port our harness to the test-matrix benchmark (lengths × channels × CFO × SNR)	Benchmark scripts and QA tests
Weeks 3–5	Prototype options 1 and 2; measure PDR and CPU on the matrix	Comparison report; mentor decision (week 4)
Week 6	Implement the chosen fix in C++ with QA tests; remove `expectedFailure`	Receiver fix with tests (midterm goal)
Weeks 7–8	Loud failures (5.2); new issue + fix/limit for ≥ 1020 B	Clear errors instead of freezes, with tests
Weeks 9–10	CSI message port + dataset example (5.3)	CSI output and dataset example
Week 11	CSI validation; tutorial page	Tutorial page; blog post
Week 12	Review changes; final report	Final evaluation

7. Expected deliverables (planned for GSoC 2027)

1. The multi-packet regression test from our existing PR #8271, finished with the maintainer, and a benchmark/test matrix.
2. A receiver-side fix for #6891 with tests and measurements; `ofdm_loopback.grc` works with the default rolloff.
3. Errors or graceful drops instead of silent freezes; resolution of the ≥ 1020 B refusal. 4.

Per-packet CSI output on `ofdm_rx` and a CSI dataset example.

5. Tutorial documentation and a final report.

8. Risks and mitigation

Risk	Mitigation
Maintainers do not want a behaviour change in `header_payload_demux`	Option 2 (timing refinement) leaves the demux unchanged; decide by week 4
C++ changes are slow to review	Small, separate changes, each with tests and numbers; tests first
CSI output API debate	Start as an example or out-of-tree block; propose in tree only after feedback
GNU Radio 4.0 priorities	Keep tests runtime-agnostic; note which results transfer to GR4
No hardware	All acceptance criteria are simulation-based; hardware is a stretch goal

9. Stretch goals

Repeat the reliability matrix on SDR hardware (PlutoSDR/USRP) or in GNU Radio's hardware in-the-loop CI.

Port the multi-packet test and CSI example to GNU Radio 4.0.

Apply the trigger-logging method to gr-ieee802-11.

10. About the applicant (each applicant fills this in)

Name, Ahmedabad University, B.Tech programme and year, time zone IST (UTC+5:30)

Availability: hours per week, exam dates, other commitments

Relevant courses: ECE 310 Wireless Communications, Signals and Systems, … Programming experience (Python, C++), GitHub profile, open-source activity (link PR #8271 if you were involved)

Why GNU Radio, and why this project

11. Communication plan

Weekly mentor call; progress on the GNU Radio Matrix chat and Discourse; a public development blog; all work done in public on GitHub.